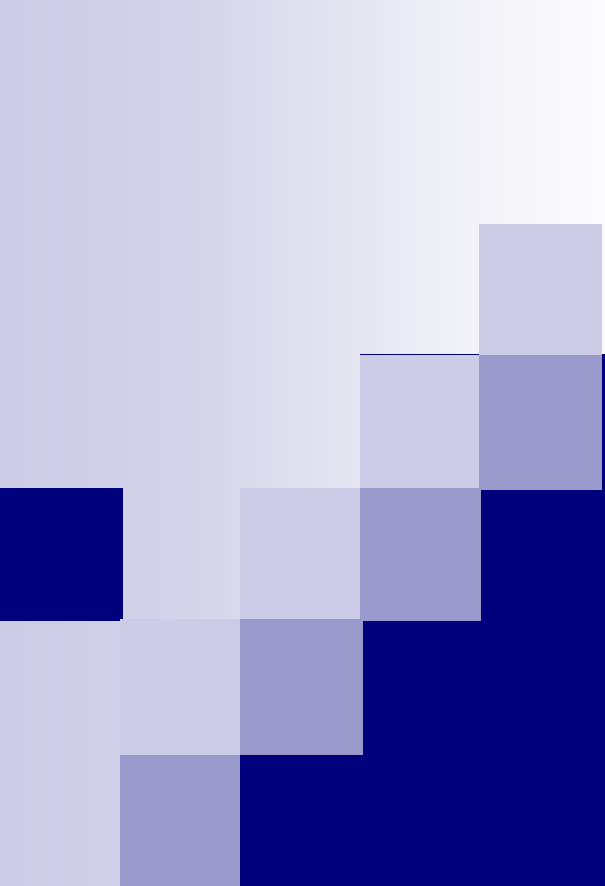




Modern Operating Systems

Chapter 1 – Introduction-1

(1.1-1.4)

Zhang Yang

cszyang@scut.edu.cn

Spring 2025



Contents of this lecture

- 1.1 What is an Operating System?
- 1.2 History of Operating Systems
- 1.3 Computer Hardware Review
- 1.4 The Operating System Zoo

- 
- 1.1 What is an Operating System?
 - 1.2 History of Operating Systems
 - 1.3 Computer Hardware Review
 - 1.4 The Operating System Zoo



What is an Operating System?(1)

- A modern computer consists of
 - One or more processors
 - Main memory
 - Disks
 - Printers
 - Various input/output devices
- Managing all these components requires a layer of software – the operating system.

What is an Operating System ? (2)

■ An Informal Definition of OS

- The OS is a collection of one or more software modules that manages and controls the resources of a computer or other computing or electronic device, and gives users and programs an interface to utilize these resources.

What is an Operating System ? (3)

■ An Informal Definition of OS (ctd.)

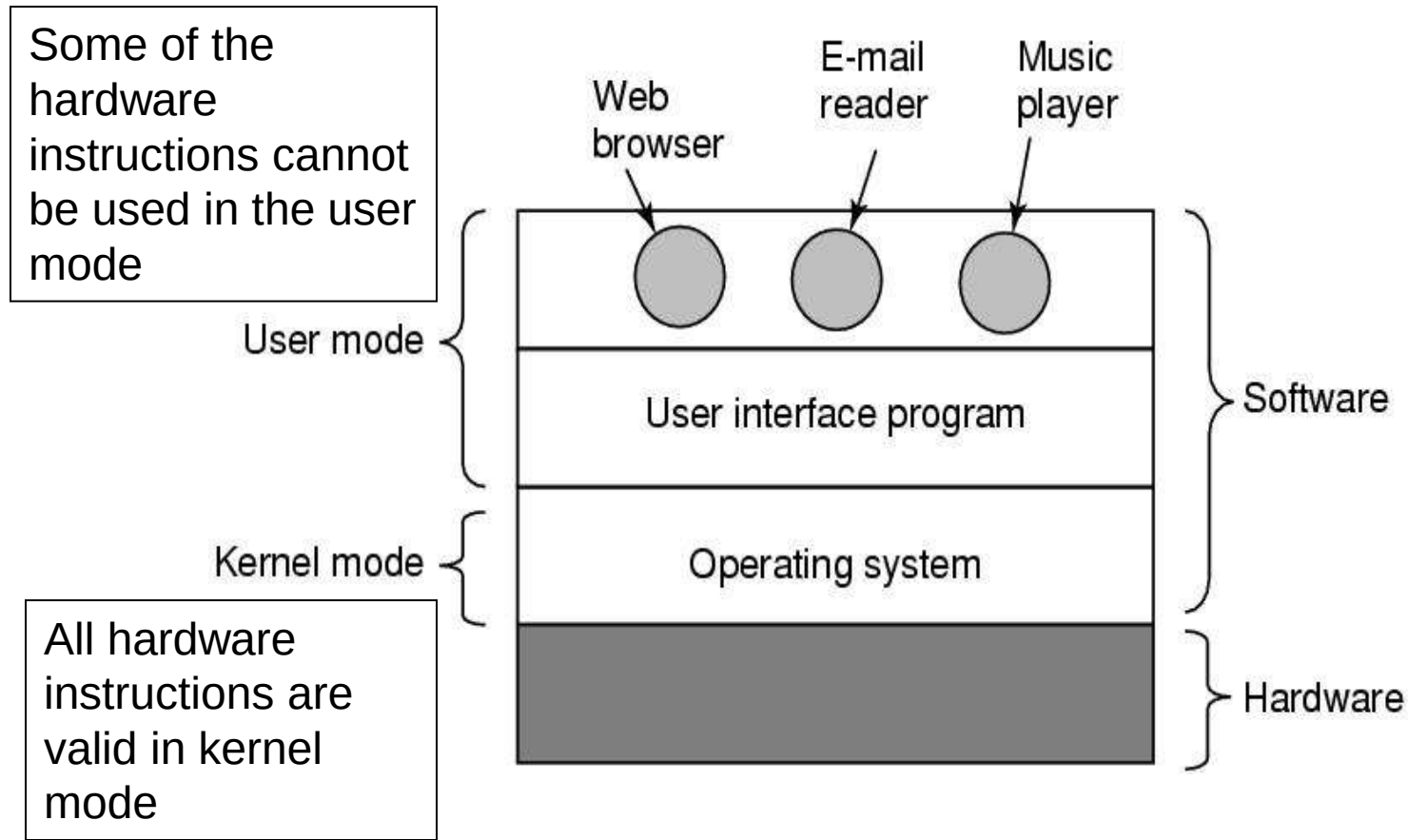


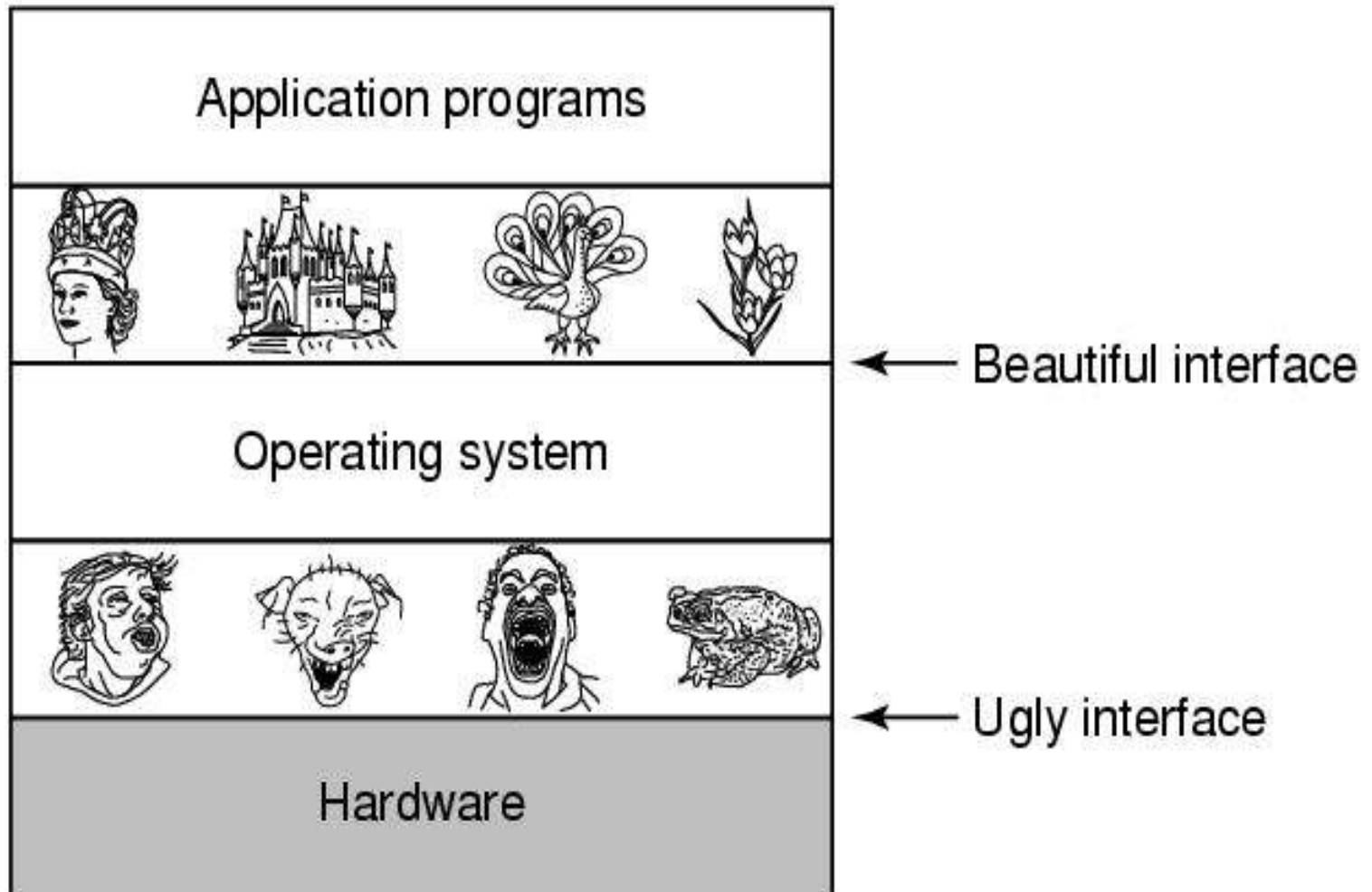
Figure 1-1. Where the operating system fits in.

What is an Operating System?(4)

- The Operating System as an Extended Machine
 - The OS hides the messy details which must be performed--Provide standard Library (i.e., abstract resources)
 - The OS implements a virtual machine on top of the machine that is easier to use (i.e. write programs for)
 - This is the “top down” (or external view) of an operating system.

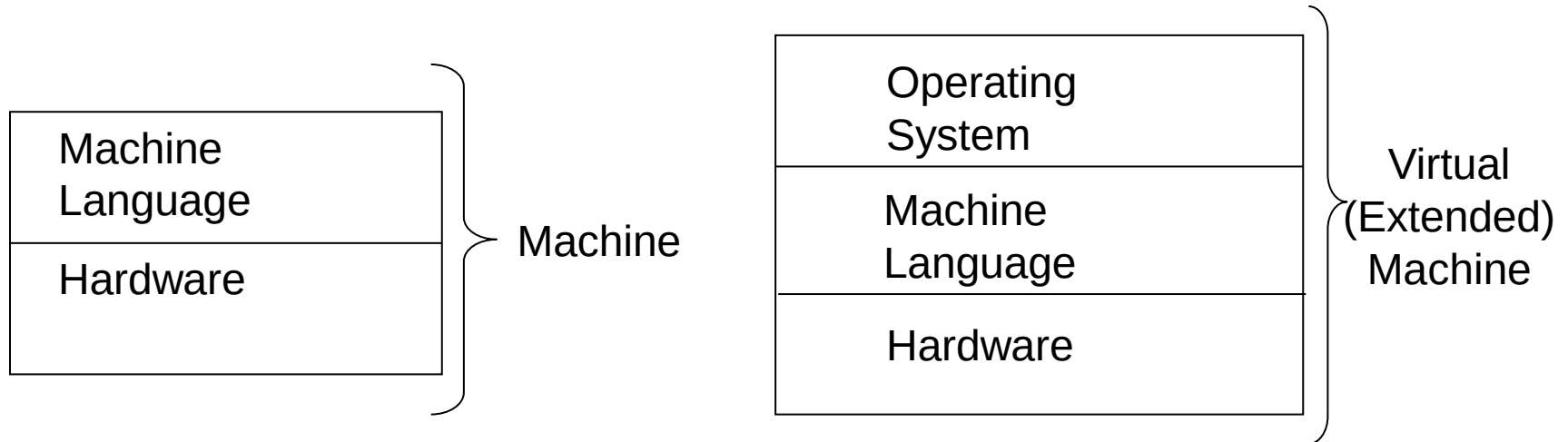
What is an Operating System?(5)

- The Operating System as an Extended Machine (ctd.)



What is an Operating System?(6)

- The Operating System as an Extended Machine (ctd.)



What is an Operating System?(7)

■ The Operating System as an Extended Machine (ctd.)

□ Example

- In a system where there is no OS installed, we should consider some hardware work as:

(Assuming an MC 6800 computer hardware)

LDAA \$80 → Loading the number at memory location 80

LDAB \$81 → Loading the number at memory location 81

ADDB → Adding these two numbers

STAA \$55 → Storing the sum to memory location 55

- As seen, we considered memory locations and used our hardware knowledge of the system.

What is an Operating System?(8)

■ The Operating System as an Extended Machine (ctd.)

□ Example (ctd.)

- In an OS installed machine, since we have an intermediate layer, our programs obtain *some advantage of mobility* by not dealing with hardware.
- For example, the above program segment would not work for an 8086 machine, where as the
“c = a + b ;”
syntax will be suitable for both.

What is an Operating System? (9)

■ The Operating System as a Resource Manager

- Manages and protects multiple computer resources: CPU, Processes, Internal/External memory, Tasks, Applications, Users, Communication channels, etc...
- Handles and allocates resources to multiple users or multiple programs running at the same time and space (e.g., processor time, memory, I/O devices).
- Decides between conflicting requests for efficient and fair resource use (e.g., maximize throughput, minimize response time).
- This is the “bottom-up” (or internal) view of an OS



What Is an OS? Real life example?

- 2-3 persons group discussion
- 2 minutes

- 
- 1.1 What is an Operating System?
 - 1.2 History of Operating Systems
 - 1.3 Computer Hardware Review
 - 1.4 The Operating System Zoo



History of Operating Systems (1)

- Two distinct phases of history
 - Phase 1: Computers are expensive
 - Goal: Use computer's time efficiently
 - Maximize throughput (i.e., jobs per second)
 - Maximize utilization (i.e., percentage busy)
 - Phase 2: Computers are inexpensive
 - Goal: Use people's time efficiently
 - Minimize response time



History of Operating Systems (2)

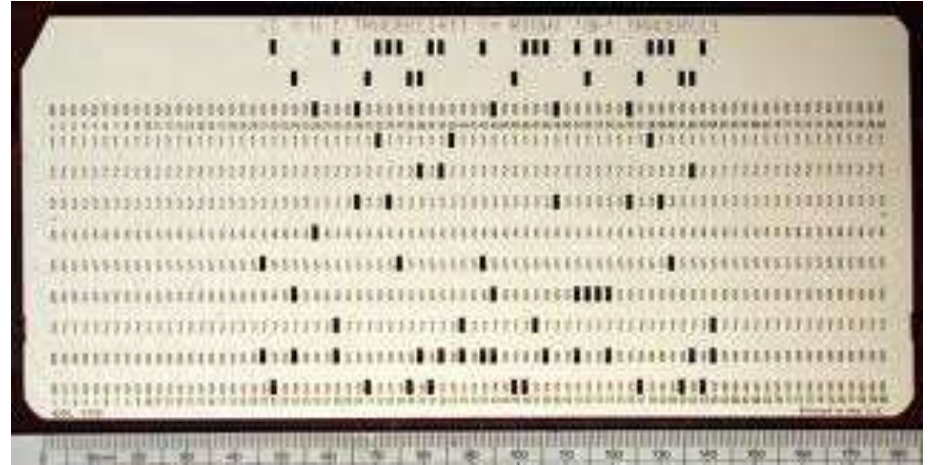
- First Generation 1945 – 1955
 - Vacuum tubes, plug boards, punched cards (no OS)
- Second Generation 1955 – 1965
 - Transistors, batch systems
- Third Generation 1965 – 1980
 - ICs, multiprogramming, spooling, timesharing
- Fourth Generation 1980 – present
 - Large scale integration
 - Personal computers, hand-held devices, sensors
- Fifth Generation 1990 – present
 - Mobile computers



History of Operating Systems (3)

- First Generation 1945 – 1955: direct input
 - Vacuum tubes, plug boards
 - Machine language, no OS
 - Punched cards(by the early 1950s)
 - User has complete control of the machine
 - Utilization is very low; set up time is high

History of Operating Systems (4)



History of Operating Systems (5)

■ Second Generation 1955 – 1965

- Transistors, batch systems
- Motivation: Increase CPU utilization
- Batch: Group of jobs submitted together
- Use a slow computer to aggregate programs and feed into fast computer.
- Same on the output side.
- Operator collects jobs; orders efficiently; runs one at a time
- Mainframe, job (a program or set of programs), FORTRAN or assembler, punched cards
- FMS (the Fortran Monitor System, IBM's OS for the 709), IBSYS (IBM's OS for the 7090 and 7094), UMES (The University of Michigan Executive System, for IBM 7094)

History of Operating System (6)

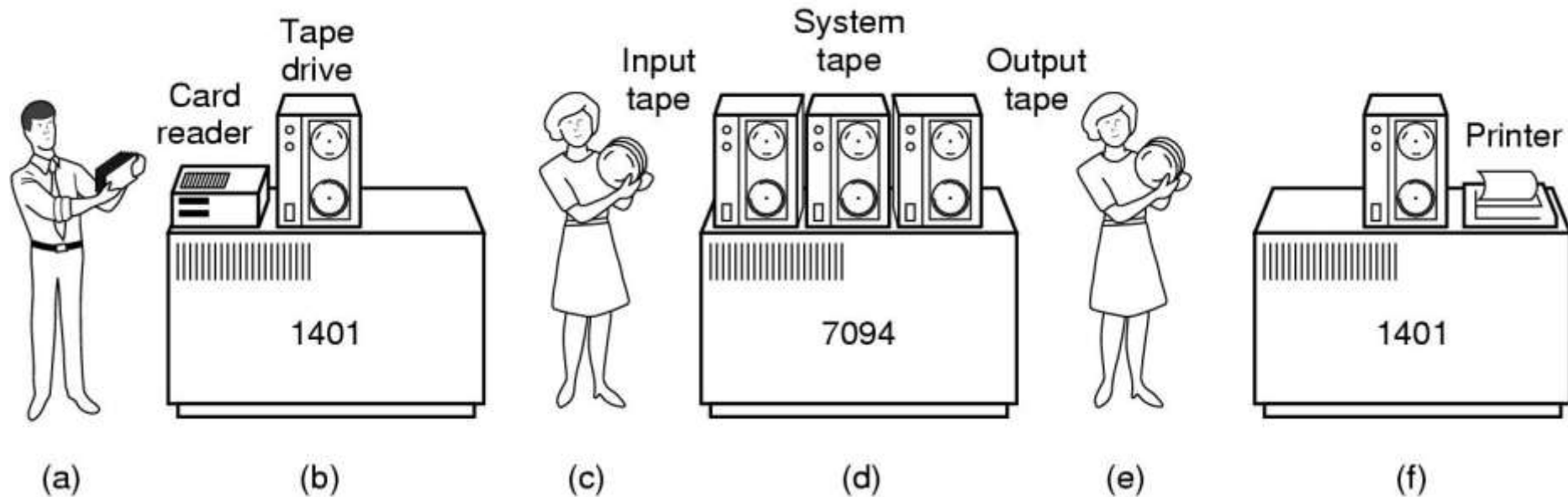


Figure 1-3. An early batch system

- (a) Programmers bring cards to 1401
- (b) 1401 reads batch of jobs onto tape
- (c) Operator carries input tape to 7094
- (d) 7094 does computing
- (e) Operator carries output tape to 1401
- (f) 1401 prints output

History of Operating Systems (7)

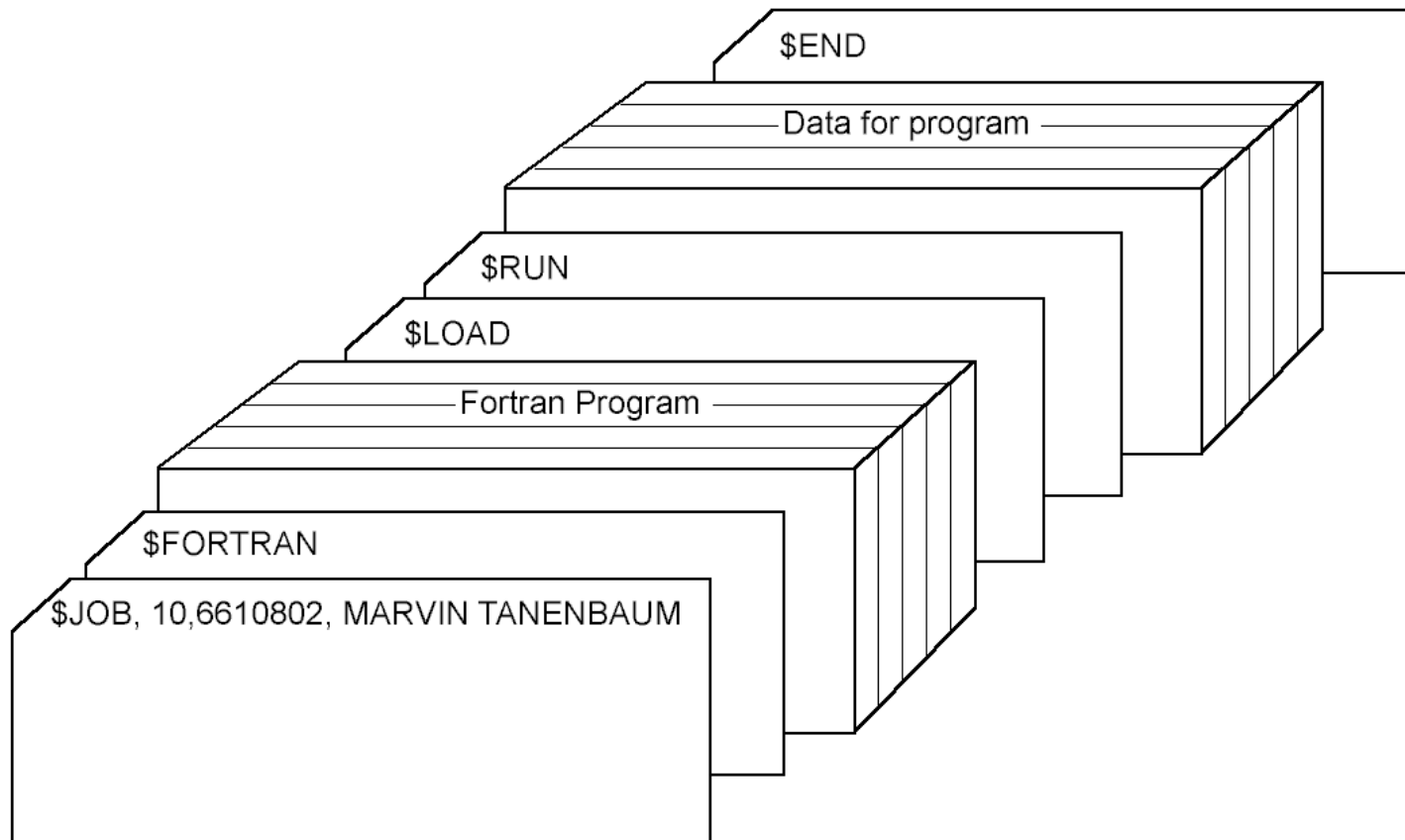


Figure 1-4. Structure of a typical FMS job

History of Operating System (8)

■ Second Generation 1955 – 1965 (ctd.)

□ Advantages

- Amortize setup costs over many jobs.
- Operator more skilled at loading tapes.
- Keep machine busy while programmer thinks.
- Improves throughput and utilization.

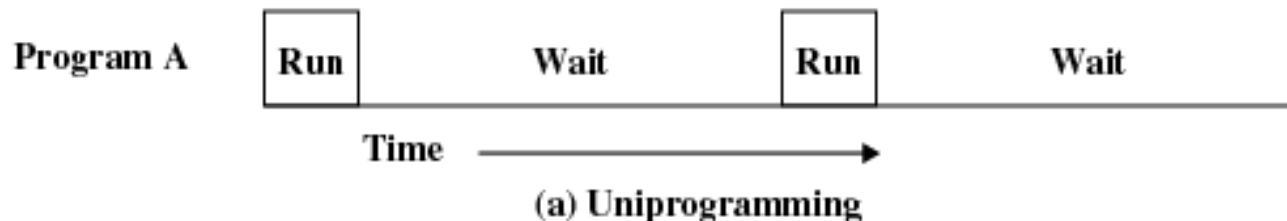
□ Problems

- User must wait until batch is done for results.
- Machine idle when job is reading from cards and writing to printers.

History of Operating Systems (9)

■ Third Generation 1965 – 1980

- ICs, multiprogramming, spooling, timesharing
- A family of software compatible computers, IBM 360 series, OS/360 (M) for IBM System 360, System 370 and System 4300. → OS/390
- Uniprogramming
 - Processor must wait for I/O instruction to complete before preceding

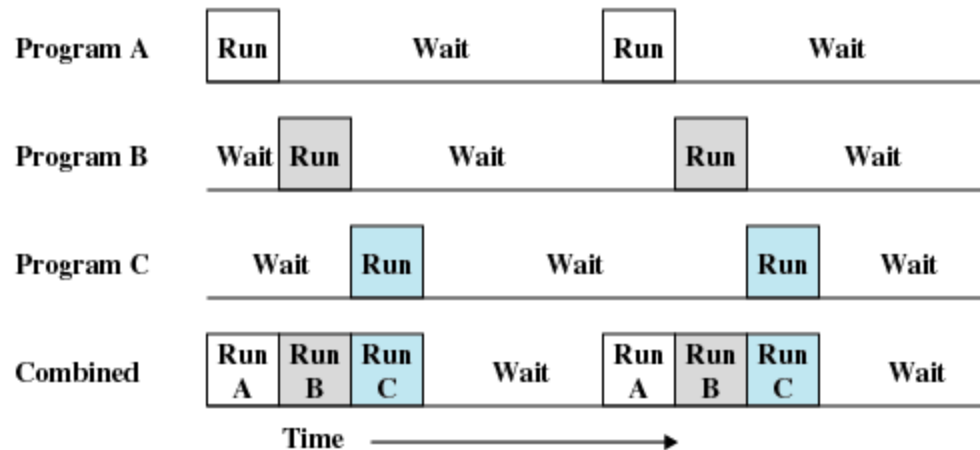


History of Operating Systems (10)

■ Third Generation 1965 – 1980 (ctd.)

□ Multiprogramming

- When one program needs to wait for I/O, the processor can switch to the other program



(c) Multiprogramming with three programs

History of Operating Systems (11)

■ Third Generation 1965 – 1980 (ctd.)

□ Multiprogramming (ctd.)

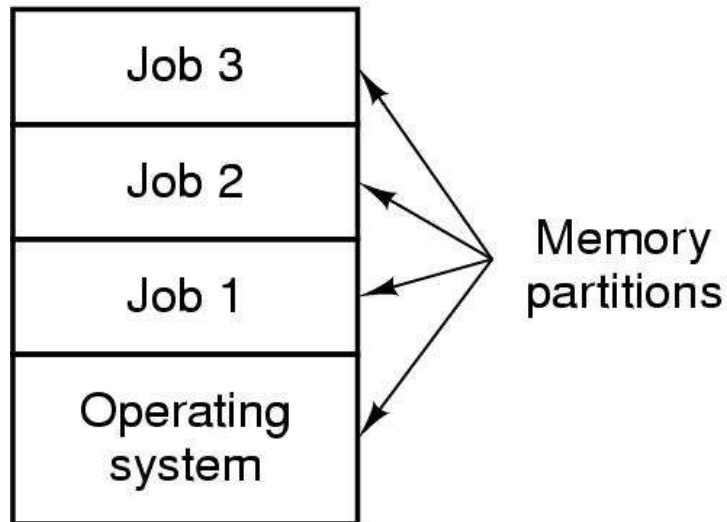


Figure 1-5. A multiprogramming system with three jobs in memory

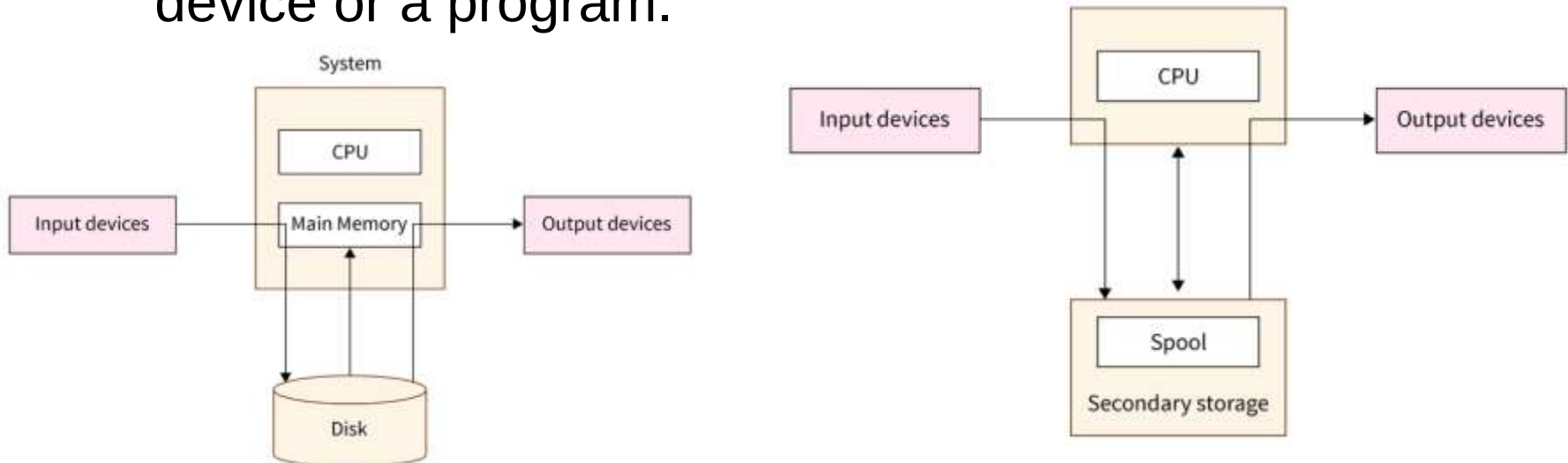
- Multiple jobs in memory
 - Protected from one another.
- Operating system protected from each job as well.
- Resources (time, hardware) split between jobs.
- Still not interactive
 - User submits job.
 - Computer runs it.
 - User gets results minutes (hours, days) later.

History of Operating Systems (12)

■ Third Generation 1965 – 1980 (ctd.)

□ Spooling

- Simultaneous Peripheral Operation On Line
- An I/O management or buffer management technique that allows the data of the input/output processes to be temporarily stored in the secondary memory which will be executed by the CPU or a device or a program.



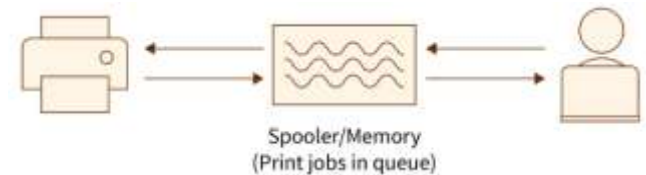
History of Operating Systems (13)

■ Third Generation 1965 – 1980 (ctd.)

□ Spooling (ctd.)

■ Application of Spooling

- Original batch systems used tape drives. Later batch systems used disks for buffering.
 - Operator read cards onto disk attached to the computer.
 - Computer read jobs from disk.
 - Computer wrote job results to disk.
 - Operator directed that job results be printed from disk.
- Use in printers. In the printing process, the documents are first stored in a queue or spool which is also known as the printer spooler. According to the process of spooling, the documents are fetched from the spool for printing once the printer is ready.



History of Operating Systems (14)

■ Third Generation 1965 – 1980 (ctd.)

□ Spooling (ctd.)

■ Advantages

- Better utilization of the expensive CPU.
- Computer overlapped I/O of one job with execution of another.

History of Operating Systems (15)

■ Third Generation 1965 – 1980 (ctd.)

□ Timesharing

- Using multiprogramming to handle multiple interactive jobs.
- Processor's time is shared among multiple users.
- Multiple users simultaneously access the system through terminals.

□ Compatible Time-Sharing System (CTSS)

- First general-purpose time-sharing system developed at MIT.

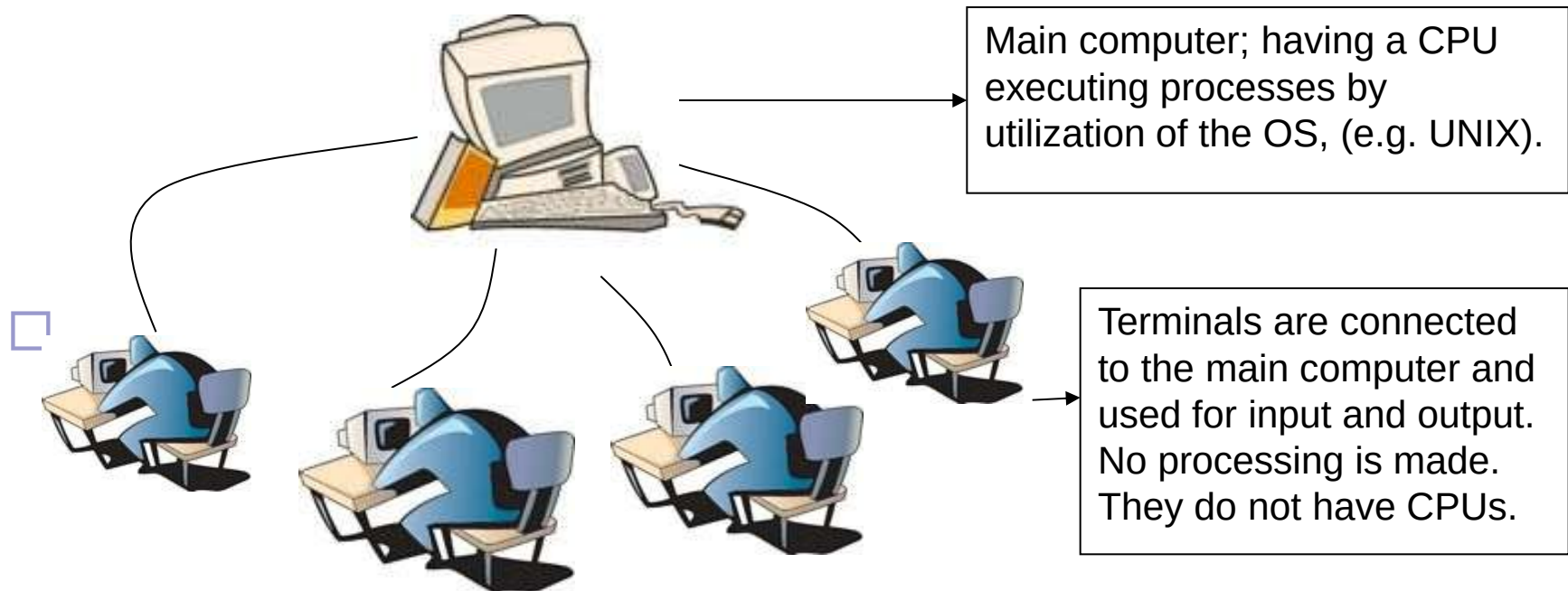
□ MULTiplexed Information and Computing Services (MULTICS)

- Developed by MIT, Bell Labs and General Electric
- Sold by Honeywell company.

History of Operating Systems (16)

■ Third Generation 1965 – 1980 (ctd.)

□ Timesharing (ctd.)



History of Operating Systems (17)

■ Third Generation 1965 – 1980 (ctd.)

□ Unix

- Ken Thompson wanted to use an old PDP-7 lying around Bell Labs.
- He and Dennis Ritchie built a system designed by programmers for programmers.
- Originally in assembly language. Rewritten by Ritchie and Thompson in C.
- New idea: portable operating system!
- Universities obtained code for experimentation.
- Berkeley added virtual memory support for the VAX.

History of Operating Systems (18)

- Third Generation 1965 – 1980 (ctd.)
 - Unix (ctd.)
 - DARPA selected UNIX as its networking platform (arpanet).
 - UNIX becomes a commercial operating system.
 - Important ideas popularized by UNIX
 - OS written in a high-level language.
 - OS is portable across hardware platforms.
 - Pipes
 - Mountable file systems.
 - Two Major Versions
 - System V (by AT&T)
 - BSD (by UC Berkely)

History of Operating Systems (19)

■ Third Generation 1965 – 1980 (ctd.)

□ MINIX

- Early version developed by Andrew S. Tanenbaum in 1987.
- A small clone of UNIX, for educational purposes.
- Since version 2.0, it has been Portable Operating System Interface compliant.
- Starting with MINIX 3, the primary aim of development shifted from education to the creation of a highly reliable and self-healing **microkernel OS**.
- MINIX 3 was developed as open-source software.

History of Operating Systems (20)

■ Third Generation 1965 – 1980 (ctd.)

□ Linux

- Linux is a family of open-source Unix-like operating systems based on the Linux kernel.
- Linux kernel began in 1991 as a personal project by Finnish student Linus Torvalds: to create a new free operating system kernel.
- Linux is typically packaged as a Linux distribution, which includes the kernel and supporting system software and libraries.

History of Operating Systems (21)

■ Fourth Generation 1980 – present

□ Windows History List

- CP/M (Control Program for Micro computer) was the first PC Operating System. 1974
- DOS (Disk Operating System) later to be renamed MS-DOS dominated the PC Market. (Windows 1.0) 1985
- Windows 3.1 provided GUI for DOS (up to Windows 3.x). 1992
- Windows 95 was first free standing version of Windows.
- Windows NT was first 32-bit OS. 1993
- Windows 95 1997
- Windows 98 1999

History of Operating Systems (22)

- Fourth Generation 1980 – present (ctd.)
 - Windows History List (ctd.)
 - Windows 2000 was introduced as Windows NT version 4. Windows Me 2000
 - Windows XP 2001
 - Windows VISTA 2007
 - Windows 7 2009
 - Windows 8 2012
 - Windows 8.1 2013
 - Windows 10 2015
 - Windows 11 2021

History of Operating Systems (23)

- Fifth Generation 1990 – present
 - Combining telephony and computing in a phone-like device since the 1970s.
 - The first real smartphone: Nokia N9000, mid-1990s
 - Symbian OS
 - RIM's Blackberry OS introduced for 2002.
 - Apple's iOS released for the first iPhone in 2007.
 - Google's Android in 2008.
 - Huawei Harmony OS 2019

- 
- 1.1 What is an Operating System?
 - 1.2 History of Operating Systems
 - 1.3 Computer Hardware Review
 - 1.4 The Operating System Zoo

A Typical Computer System

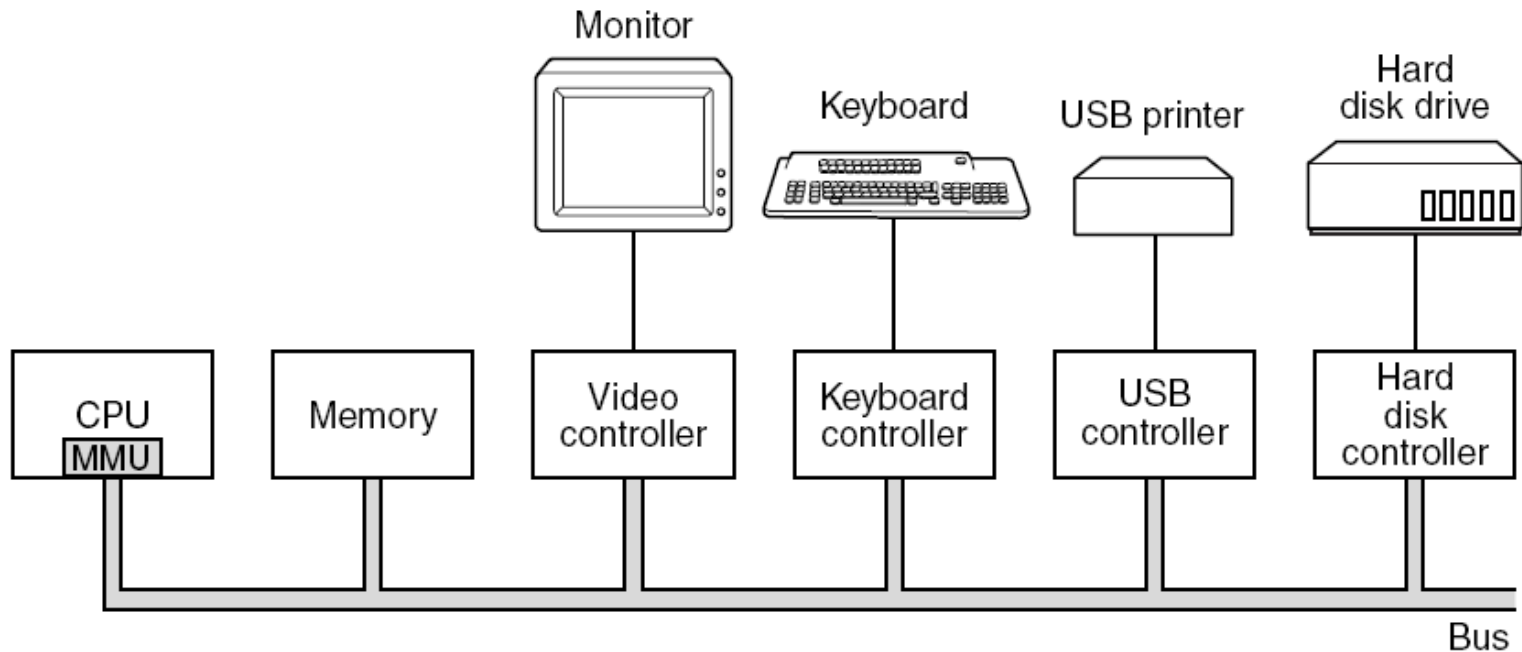


Figure 1-6. Some of the components of a simple personal computer.

Processors

- Each CPU has a specific set of instructions.
- All CPUs contain
 - General registers inside to hold key variables and temporary results.
 - Special registers visible to the programmer.
 - Program counter contains the memory address of the next instruction to be fetched.
 - Stack pointer points to the top of the current stack in memory.
 - PSW (Program Status Word) contains the condition code bits which are set by comparison instructions, the CPU priority, the mode (user or kernel) and various other control bits.



How Processors Work (1)

■ Execute Instructions

□ CPU Cycles

- Fetch (from mem) → decode → execute
- Program Counter (PC)
 - When is PC changed?
- Pipeline: fetch $n+2$ while decode $n+1$ while execute n
- Superscalar

How Processors Work (2)

■ Execute Instructions (ctd.)

□ CPU Cycles (ctd.)

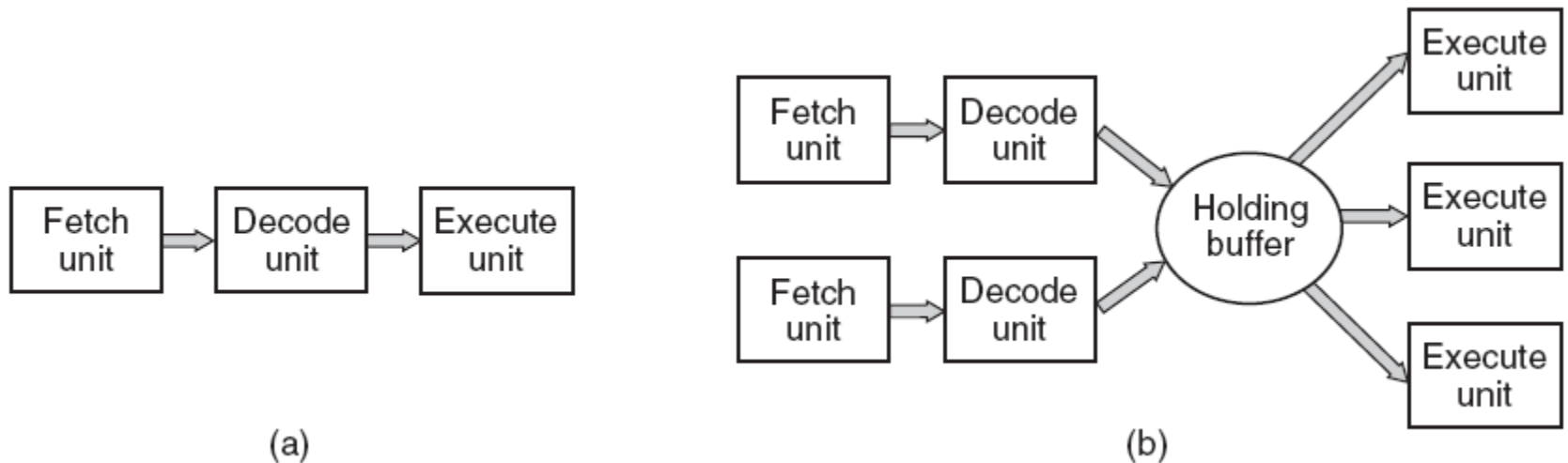


Figure 1-7. (a) A three-stage pipeline. (b) A superscalar CPU.



How Processors Work (3)

- Execute Instructions (ctd.)
 - Two Modes of CPU (why?)
 - User mode (a subset of instructions)
 - Kernel mode (all instruction)
 - Trap (special instruction)
 - Switch from user mode to kernel mode and starts the OS.

Multithreaded and Multicore Chips

■ Multithreading or Hyperthreading

- Allow the CPU to hold the state of two different threads and then switch back and forth on a nanosecond time scale.

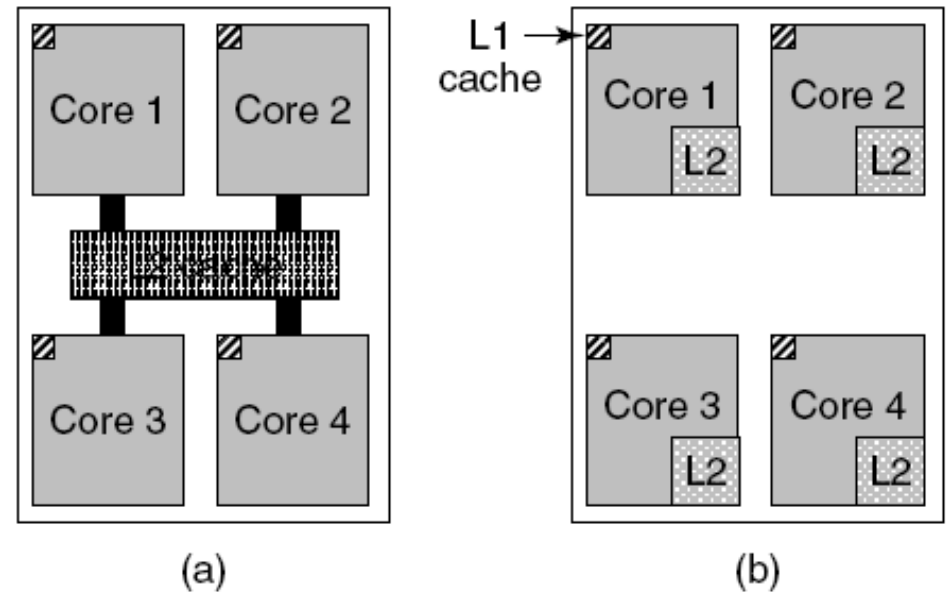


Figure 1-8. (a) A quad-core chip with a shared L2 cache. (Intel)
(b) A quad-core chip with separate L2 caches. (AMD)

Memory-Storage Hierarchy

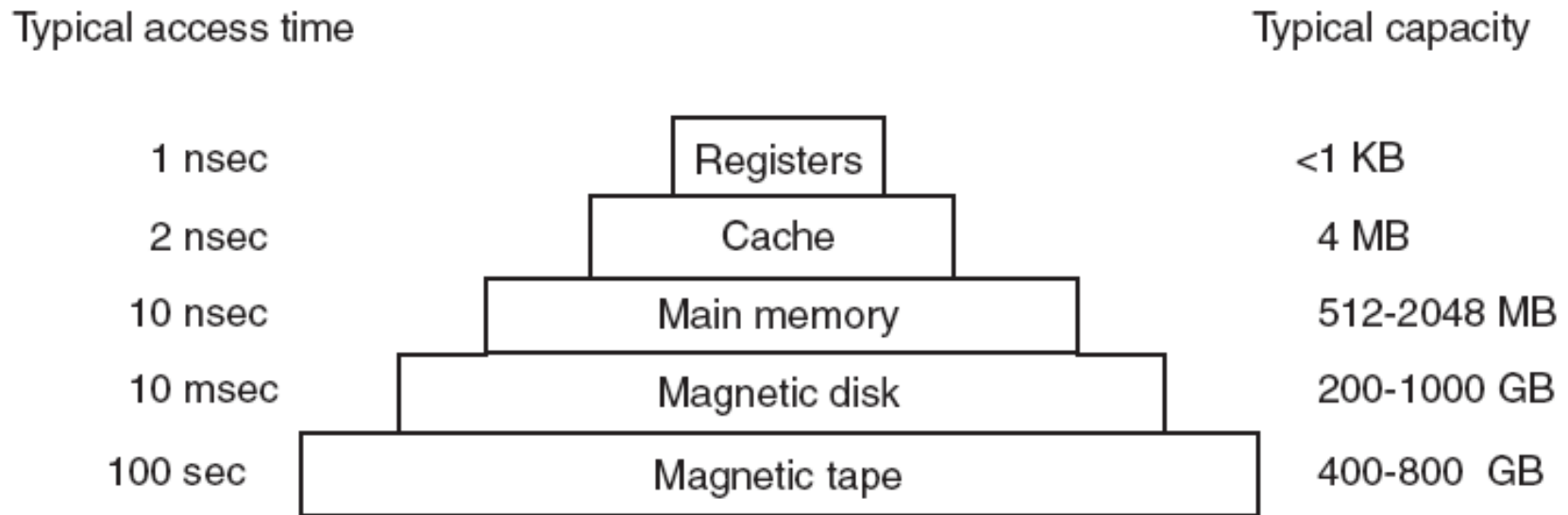
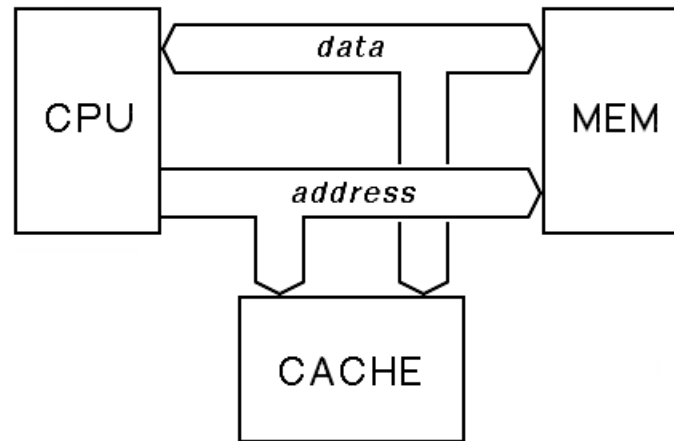


Figure 1-9. A typical memory hierarchy.

The numbers are very rough approximations.

Real life analogy?

Cache (1)



- Cache Hit

- ☐ No need to access memory

- Cache Miss

- ☐ Data obtained from memory, possibly update cache



Cache (2)

- Questions when dealing with cache
 - When to put a new item into the cache.
 - Which cache line to put the new item in.
 - Which item to remove from the cache when a slot is needed.
 - Where to put a newly evicted item in the larger memory.

Disks

- One or more metal platters
- 5400/7200/10800 rpm
- Track
- Cylinder
- Sector
- Virtual memory

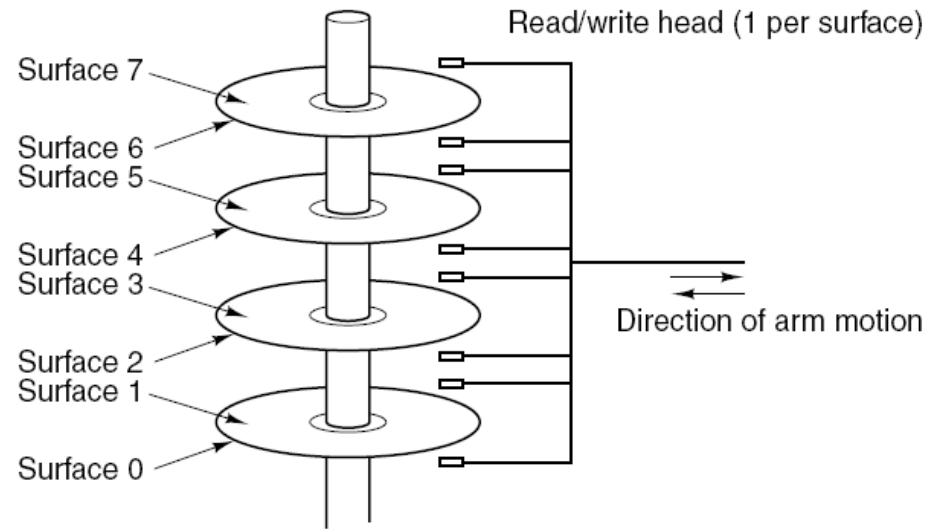


Figure 1-10. Structure of a disk drive.



I/O Devices (1)

■ Controller

- Example: Disk Controller
- Controllers are complex converting OS request into device parameters.
- Controllers often contain small embedded computers.

■ Device

- Fairly simple interfaces and standardized.
- SATA (Serial AT Attachment) – standard disk type on many computers.

I/O Devices (2)

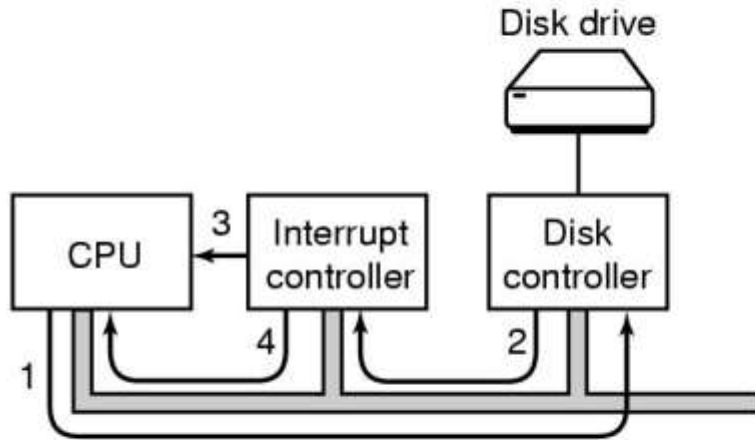
■ Device Driver

- Software that talks to a controller, giving it commands and accepting responses.
- Needed since each type of controller may be different.
- It has to be put into the OS so it can run in kernel mode.
- Each controller manufacturer supplies a driver for each OS it supports (e.g., drivers for Windows XP, Longhorn, UNIX).

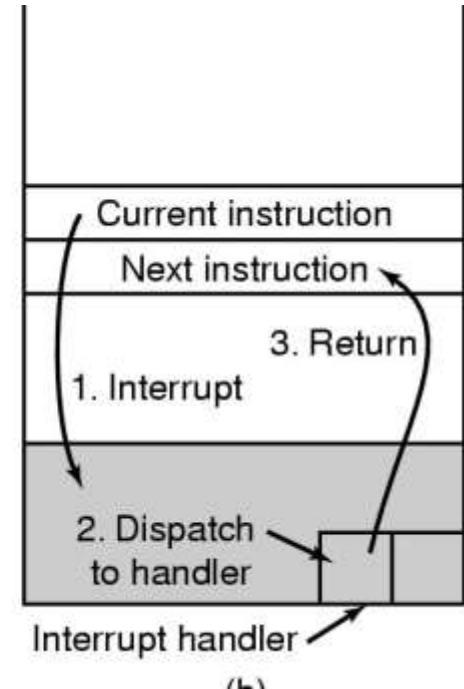
Methods for I/O (1)

- How device driver talks to controller.
 - Busy waiting
 - CPU executes a program that communicates with controllers.
 - Read/write commands
 - Sensing status
 - Transferring data
 - CPU waits for controllers to complete operation.
 - Disadvantage is that it wastes CPU time.
 - Interrupt
 - DMA

Methods for I/O (2)



(a)



(b)

Figure 1-11.

- (a) The steps in starting an I/O device and getting interrupt.
- (b) Interrupt processing involves taking the interrupt, running the interrupt handler, and returning to the user program.

Buses

- The OS must be aware of all of buses for configuration and management.

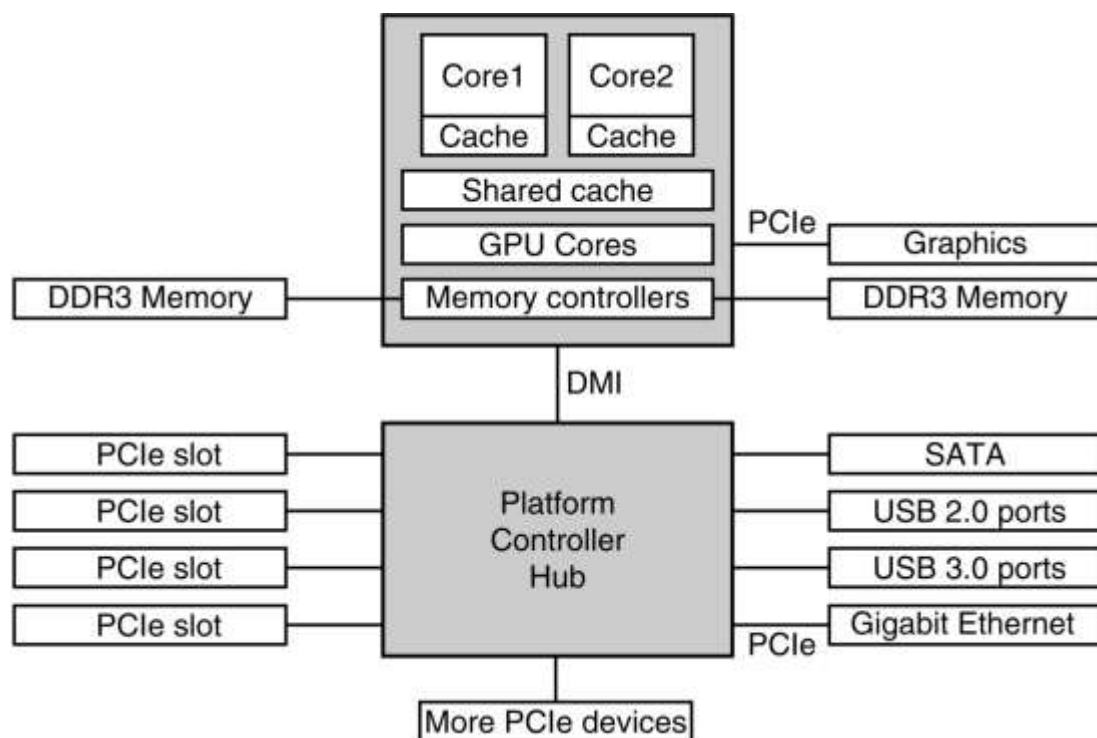


Figure 1-12. The structure of a large x86 system

- 
- 1.1 What is an Operating System?
 - 1.2 History of Operating Systems
 - 1.3 Computer Hardware Review
 - 1.4 The Operating System Zoo

The Operating System Zoo (1)

■ Mainframe Operating Systems

- Very powerful systems used on large computers (e.g. in insurance companies, banks)
- Support batch processing, transaction processing, and timesharing
- OS/390, Linux

■ Server Operating Systems

- Large PC's that provide services (usually over a network), e.g. print service, file service or Web service
- Solaris, FreeBSD, Linux or Windows Server 200x

The Operating System Zoo (2)

■ Multiprocessor Operating Systems

- Special versions of server OS for multiprocessor hardware.
- HYDRA and StarOS developed by CMU.
 - Object-oriented Operating System
- Unix-like Operating Systems
 - DYNIX (Dynamic Unix)
 - Developed by Sequent Computer Systems.
 - Based on 4.2BSD and modified to run on Intel-based symmetric multiprocessor hardware.
 - UMAX, RP3
- Popular OS
 - Windows
 - Linux
 - ...



The Operating System Zoo (3)

- Personal Computer Operating Systems
 - Windows, Linux, FreeBSD, MacOS
- Handheld Computer Operating Systems
 - Symbian OS, Palm OS, Android, iOS
- Embedded Operating Systems
 - No untrusted software will run on embedded systems.
 - QNX, Vxworks



The Operating System Zoo (4)

- Sensor Node Operating Systems
 - TinyOS
- Real-Time Operating Systems
 - Special OS that operates in real time
 - Response of hardware must meet strict time requirements
 - e-Cos
- Smart Card Operating Systems
 - The OS is the hardware-specific firmware that provides basic functionality as secure access to on-card storage, authentication and encryption.
 - Only a few cards allow writing programs that are loaded onto the smart card.